

The Recurrence of $2/3$:

**Koide Projectors, Complement Step,
FFGFT Ladder and $D = 3 + \varepsilon$ Deformation**

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Abstract

At exactly $D = 3$, three distinct mathematical structures yield the value $2/3$: the Koide equal-projector condition $A = B$ between democratic and orthogonal projection components, the complement step $1 - 1/D$, and the geometric FFGFT lepton ladder with ratio $q = 2/3$. A $D = 3 + \varepsilon$ deformation separates the three roads. This document shows: (i) the FFGFT ladder is geometric, not arithmetic; (ii) the exponent step is tied to the integer topological $D = 3$, not to the fractal dimension D_f ; (iii) without Koide input the FFGFT ladder gives $Q \approx 0.6677$ (0.16 % above $2/3$) – since r_i and p_i are adjusted to the measured masses, Koide follows from the ladder only approximately and is no independent confirmation of the exponent structure.

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Notation

Symbol	Meaning
$\xi = 4/30000$	Fundamental FFGFT parameter [7]
$v = 246.22 \text{ GeV}$	Higgs vacuum expectation value
$m_i = r_i \xi^{p_i} v$	FFGFT mass formula [4]
r_i	Rational prefactors of the leptons
p_i	Rational exponents of the leptons [5]
$D = 3$	Topological (integer) spatial dimension of T^4
$D_f^{\text{space}} = 3 - \xi$	Fractal spatial dimension (local metric) [7]
$D_f^{\text{eff}} \approx 2.973$	Effective fractal dimension (scale flow)
K_{frak}	Fractal correction of absolute scales (value $1 - 100\xi$, form open)
$x_i = \sqrt{m_i}$	Square-root lepton mass
$u = (1, 1, 1)^T$	Democratic unit vector
$P = uu^T/3$	Democratic projector
$P^\perp = I - P$	Residual projector
$A = x^T P x$	Democratic quadratic component
$B = x^T P^\perp x$	Residual quadratic component
Q	Koide quantity $= (A + B)/(3A)$ [3]
$\Delta_K = 3Q - 2$	Koide residual diagnostic
ξ_{Koide}	Value of ξ for exact $Q = 2/3$
α	Fine-structure constant

.1 Starting Point: Three Roads to $2/3$

At exactly $D = 3$, several independent principles produce the same numerical value:

$$1 - \frac{1}{D} = \frac{2}{3}, \quad \frac{2}{D} = \frac{2}{3}, \quad \frac{1}{2} \left(1 + \frac{1}{D} \right) = \frac{2}{3}. \quad (.1)$$

The useful instrument for distinguishing them is the formal deformation $D = 3 + \varepsilon$: principles that coincide at $D = 3$ separate under this continuation.

The following table summarises the three continuations:

Principle	Formula	Expansion at $D = 3 + \varepsilon$
Koide equal-projector	$\frac{2}{D}$	$\frac{2}{3} - \frac{2\varepsilon}{9} + O(\varepsilon^2)$
Complement step	$1 - \frac{1}{D}$	$\frac{2}{3} + \frac{\varepsilon}{9} + O(\varepsilon^2)$
Interval midpoint	$\frac{1}{2} \left(1 + \frac{1}{D} \right)$	$\frac{2}{3} - \frac{\varepsilon}{18} + O(\varepsilon^2)$

The roads separate at first order – the complement step rises, the Koide equilibrium road and the interval midpoint fall.

.2 The Koide Relation as a Projector Statement

For the charged leptons define $x_i = \sqrt{m_i}$ and write the Koide quantity as

$$Q = \frac{\sum_i m_i}{(\sum_i \sqrt{m_i})^2} = \frac{x^T x}{(u^T x)^2}, \quad u = (1, 1, 1)^T. \quad (.2)$$

With the democratic projector $P = uu^T/3$ and residual projector $P^\perp = I - P$:

$$A = x^T P x = \frac{(u^T x)^2}{3}, \quad B = x^T P^\perp x, \quad Q = \frac{A + B}{3A}.$$

The Koide value $Q = 2/3$ is equivalent to the **equal-projector condition** $A = B$: the square-root mass vector splits into a democratic and a residual component of equal quadratic weight.

Under formal D -dimensional extension with $Q_D = 2/D$:

$$Q_{3+\varepsilon} = \frac{2}{3+\varepsilon} = \frac{2}{3} - \frac{2\varepsilon}{9} + O(\varepsilon^2) \quad (\text{descending}). \quad (.3)$$

.3 The FFGFT Ladder is Geometric, not Arithmetic

The canonical charged-lepton parameters in FFGFT [7, 4] are:

$$(r_e, r_\mu, r_\tau) = \left(\frac{4}{3}, \frac{16}{5}, \frac{25}{9} \right), \quad (p_e, p_\mu, p_\tau) = \left(\frac{3}{2}, 1, \frac{2}{3} \right), \quad (.4)$$

with $m_i = r_i \xi^{p_i} v$, $\xi = 4/30000$, $v = 246$ GeV (Higgs vacuum value).

The exponents $(\frac{3}{2}, 1, \frac{2}{3})$ do *not* form a uniform $1/3$ ladder – the gaps are $p_e - p_\mu = 1/2$ and $p_\mu - p_\tau = 1/3$. The resolution: the ladder is **geometric** with constant ratio $q = 2/3$:

$$p_{n+1} = \frac{2}{3} p_n, \quad \frac{3}{2} \xrightarrow{\times 2/3} 1 \xrightarrow{\times 2/3} \frac{2}{3}. \quad (.5)$$

The unequal differences $1/2$ and $1/3$ are *consequences* of a constant ratio, not independent rules.

Two equivalent readings.

- **Ratio reading.** $q = 2/3$ is the inverse of the tetrahedral geometry factor $3/2$. The same number that orders the ladder is, numerically, the Koide value.
- **Complement reading.** At the $\mu \rightarrow \tau$ rung $p_\tau = 1 - 1/D = 2/3$ with $D = 3$. This places FFGFT on the complement-step road – *not* on the equal-projector road $2/D$.

The electron at $p_e = 3/2 = 1 + 1/(D-1)$ is a half-integer rung – a geometric midpoint between winding levels (the spin-winding contribution [5]), not a full winding step. That is the structural origin of the $1/2$ gap.

.4 Topological $D = 3$ versus the Fractal Dimension

FFGFT carries two derived fractal dimensions:

$$D_f^{\text{space}} = 3 - \xi \approx 2.9998667 \quad (\text{local metric; GRB time-of-flight}), \quad (.6)$$

$$D_f^{\text{eff}} \approx 2.973$$

(cumulative scale flow, Planck to electroweak).

Both are derived, not assumed.

Why no $\xi/9$ correction reaches the Koide quantity. The Koide quantity Q is a pure mass *ratio*. Every scale-dependent factor – the vacuum value v , the fractal correction K_{frak} , and any D_f -level shift – is common to all three leptons and cancels in Q . Fractal corrections attach to *absolute* scales, never to ratios.

The exponent step is therefore tied to the integer, topological $D = 3$. The shift $2/3 - \xi/9$ from the D_f road is a real effect on absolute predictions; it simply does not enter Q .

.5 Forced or Merely Fitted? An Honest Answer

What the ladder fixes

The prefactors r_i and exponents p_i are settings, adjusted to the measured masses; agreements of individual masses with the measured values are therefore not

evidence. The ratios follow from the ladder to a few digits, for instance $m_\mu/m_e = \frac{12}{5} \xi^{-1/2} \approx 207.8$. There is no second, independent road via K_{frak} for this ratio: K_{frak} is common to all three leptons and cancels in ratios (Section .4).

What the ladder does not force

The ladder does not force the equal-projector condition $A = B$ exactly. Inserting the FFGFT masses into the Koide formula gives [6]:

$$Q_{\text{FFGFT}} \approx 0.6677, \quad \Delta_K = 3Q_{\text{FFGFT}} - 2 \approx +0.003 \quad (.7)$$

(0.16 % above the exact value $2/3$).

The reason is structural: the Koide denominator $(\sum \sqrt{m_i})^2$ generates cross-terms with exponents

$$\xi^{5/4}, \quad \xi^{13/12}, \quad \xi^{5/6}, \quad (.8)$$

none of which is a multiple of $1/3$. They lie *outside* the FFGFT structure space $\{p\} \subset \frac{1}{3}\mathbb{Z}$. Exact $Q = 2/3$ would require $\xi_{\text{Koide}} \approx 1.372 \times 10^{-4}$, about 2.9 % larger.

The honest position: The ladder reproduces Q to 0.16 % from (r_i, p_i, ξ) with no Koide input. Since r_i and p_i are adjusted to the measured masses, this is no independent confirmation of the exponent structure: Koide follows from the ladder only approximately and is not algebraically forced by it. The small residual lies precisely where the cross-term analysis predicts.

.6 The Explicit Map FFGFT $\rightarrow$ Koide

The three structures – the Koide projector form, the FFGFT ξ -ladder, and Paul’s 24-cell rank split¹ – are three distinct primitive objects. A relation between them must be exhibited by a map, not by a shared numeral.

The FFGFT-to-Koide map is:

$$(r_i, p_i, \xi) \mapsto x_i = \sqrt{r_i} \xi^{p_i/2} \sqrt{v} \mapsto (A, B) \mapsto Q_{\text{FFGFT}} = \frac{\sum_i r_i \xi^{p_i}}{(\sum_i \sqrt{r_i} \xi^{p_i/2})^2}. \quad (.9)$$

Since $\sqrt{v}$ cancels, the map is explicit and trigonometry-free.

The Δ_K discriminator:

$$\Delta_K = 3Q_{\text{FFGFT}} - 2 = \frac{B}{A} - 1 \approx +0.003 \quad (.10)$$

confirms that the ladder road and the equal-projector road meet near $2/3$ and separate – exactly as the ε -deformation predicts.

¹P. Phillips (IPI correspondence, May 2026 [1]) postulates a 24-dimensional mediation space with an involution J , $J^2 = I$, whose projectors $P_\pm = (I \pm J)/2$ have ranks $\text{rank}(P_-) = 16$ and $\text{rank}(P_+) = 8$. This yields the ratios $8/24 = 1/3$ and $16/24 = 2/3$ as a pure rank statement, independent of masses or exponents. The geometric motivation is the regular 24-cell polytope (24 vertices in $\mathbb{R}^4$) whose symmetry group F_4 admits a natural $8 + 16$ splitting of the weight space. A formal map to FFGFT quantities has not yet been established.

.7 Summary

1. At $D = 3$ three roads converge to $2/3$; the $D = 3 + \varepsilon$ deformation separates them.
2. The FFGFT ladder $(\frac{3}{2}, 1, \frac{2}{3})$ is geometric with ratio $q = 2/3$; the unequal gaps $1/2$ and $1/3$ are consequences.
3. FFGFT lies on the complement road $p_\tau = 1 - 1/D$, not on the equal-projector road $2/D$.
4. Fractal corrections (D_f) cancel in Q ; the step is tied to topological $D = 3$.
5. The ladder gives $Q \approx 0.6677$ (0.16 % above $2/3$) without Koide input; the residual arises from cross-term exponents outside $\frac{1}{3}\mathbb{Z}$.
6. Exact $Q = 2/3$ would require $\xi_{\text{Koide}} \approx 1.372 \times 10^{-4}$; since r_i and p_i are adjusted to the measured masses, Koide follows from the ladder only approximately.

Methodological classification: The algebraic derivations in Sections .3–.4 are bridges (directly from the ladder structure and torus topology). The connection to Paul's 24-cell rank split is a plausibility sketch; a formal map requires further elaboration.

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